

Heart Transplant, Pediatric

ORG: P-535 (ISC)

[Link to Codes](#)

MCG Health

Inpatient & Surgical

Care

28th Edition

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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **ALL** of the following(1)(2)(3)(4)(5):[N](#)
 - ☐ Transplant need, as indicated by **1 or more** of the following:
 - Cardiogenic shock or severe heart failure requiring continuous intravenous inotropic support or mechanical cardiac support (eg, intra-aortic balloon pump, extracorporeal membrane oxygenation, ventricular assist device)(10)(11)(12)(13)
 - ☐ Symptomatic heart failure and **1 or more** of the following:
 - Peak metabolic oxygen consumption[A] on cardiopulmonary exercise test less than 50% predicted based on patient age, sex, and weight
 - Growth failure attributed to heart disease
 - Recurrent symptomatic or life-threatening arrhythmia unresponsive to medical therapy and interventional procedures (eg, catheter ablation, intracardiac defibrillator)
 - Severe exercise or activity intolerance
 - Progressive pulmonary hypertension
 - Recurrent hospitalizations for heart failure
 - ☐ Severe congenital heart disease, as indicated by **1 or more** of the following(2)(15):
 - Hypoplastic left heart syndrome and **1 or more** of the following(16):
 - Proximal coronary artery stenosis or atresia
 - Atrioventricular or semilunar valve with moderate to severe stenosis or insufficiency
 - Severe ventricular dysfunction
 - Ventricular failure due to congenital heart disease that is not amenable to other surgical alternatives(17)
 - Severe arterial oxygen desaturations (cyanosis) not amenable to other surgical correction
 - Fontan circulation[B] with systemic complications, as indicated by **1 or more** of the following(18):
 - Protein losing enteropathy
 - Plastic bronchitis
 - Stroke or thromboembolic disease
 - Refractory ascites
 - Cirrhosis of the liver
 - Failed surgical palliation

☐ Low-grade myocardial tumor and **ALL** of the following:

- No evidence of metastatic disease
- Tumor is otherwise unresectable.
- Retransplant for graft dysfunction from moderate or severe allograft vasculopathy
- o No contraindications to heart transplant present

Alternatives to Procedure

- Alternatives include(1)(2)(3)(4)(5)(19)(20):[1](#)
 - o Maximum medical management of heart failure
 - o For fulminant myocarditis: corticosteroids and immunosuppressive therapy, discontinuation of causative agent (eg, drug hypersensitivity reaction, immune checkpoint inhibitor cardiotoxicity) if appropriate, or possible mechanical assist device(21)
 - o Combined cardiac resynchronization therapy and implantable cardioverter-defibrillator. See Electrophysiologic Study and Implantable Cardioverter-Defibrillator (ICD) Insertion [4](#) ISC guideline.
 - o Ventricular assist device (mechanical circulatory support) for(13)(22)(23)(24):
 - Recovery support in postcardiotomy, postinfarction, or post-transplant "stunned" myocardium, or for myocarditis
 - Bridging support in transplant or retransplant recipients in life-threatening heart failure, arrhythmias, or end organ failure
 - o Nontransplant surgical alternatives for heart failure, including(15):
 - Cardiac valve replacement(25)
 - Intra-aortic balloon pump(10)
 - Left ventricular reconstruction
 - Creation of Fontan circulation or other procedures for complex congenital heart disease
 - Total artificial heart(13)(26)
 - o Palliative care(27)(28)(29)

Operative Status Criteria

Goal Length of Stay: 12 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(1)(3)(4)(8)(19)(20):
 - o Preoperative evaluation, including:
 - Thorough evaluation by multidisciplinary heart transplant team for:
 - Severity of heart disease
 - Alternatives to heart transplant
 - Contraindications to heart transplant
 - Psychosocial evaluation (eg, mental health, social support)(30)(31)
 - Protocol as specified by transplant center
 - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool [4](#) SR.
 - o Diagnostic test scheduling, including:
 - Cardiovascular testing
 - Pulmonary testing
 - Cardiopulmonary exercise test (to measure peak metabolic oxygen consumption and rule out pulmonary cause of exercise limitation)
 - Laboratory testing
 - Nutrition evaluation
 - Dental evaluation
 - Immunology screening, including blood and tissue typing
 - Detailed infectious disease screening, including latent tuberculosis and viral antibody titers(3)
 - o Preoperative treatment, procedures, and stabilization, including:
 - Optimizing cardiac medication regimen
 - Pulmonary vasodilator agents (eg, inhaled nitric oxide, prostacyclin)(7)
 - Ventricular assist device (left, right, or biventricular mechanical circulatory support)(22)(24)(32)(33)
 - Extracorporeal membrane oxygenation as bridge to transplant or to ventricular assist device(34)
 - Pretransplant desensitization to HLA antibodies(35)
 - Addressing all treatable coexisting conditions before transplant
 - o Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	- OR to ICU - Social Determinants of Health Assessment - Discharge planning	Clinical Indications met[C] - Intubated	- Bed rest - Turn side to side every 2 hours	- IV fluids, medications - IV medications - NPO	- Ventricular assist device - Central lines - Arterial line - Pacing wires - Continuous ECG monitoring - Oximetry - CXR postoperatively - Oxygen - Chest tube	- Perioperative antibiotics - Immunosuppressive medications - Possible inhaled pulmonary vasodilators - Possible parenteral vasoactive medication
2	- ICU - Social Determinants of Health Assessment	No unexpected ECG changes	- Bed rest - Turn side to side every 2 hours	- NPO - IV fluids - IV medications	- Central lines - Continuous ECG monitoring - Temporary pacing if indicated - Oxygen - Oximetry - Chest tube - Portable CXR	- Perioperative antibiotics - Immunosuppressive medications - Possible inhaled pulmonary vasodilators - Possible parenteral vasoactive medication
3	- ICU - Social Determinants of Health Assessment	Pain absent or managed Mechanical ventilation absent	Bed rest	- Possible clear liquid diet - IV fluids - IV medications	- Oxygen - Oximetry - ECG monitoring - Temporary pacing if indicated	- Antibiotics absent - Possible inhaled pulmonary vasodilators - Possible parenteral vasoactive medication - Immunosuppressive medications - Aspirin
4	- ICU or intermediate care - Social Determinants of Health Assessment	Hemodynamic stability	Dangle legs[D]	Clear liquid diet - IV fluids - IV medications	Chest tube absent - Oxygen - Oximetry - ECG monitoring - Temporary pacing if indicated	- Immunosuppressive medications - Possible inhaled pulmonary vasodilators - Possible parenteral vasoactive medication - Aspirin
5	- ICU or intermediate care - Possible floor - Social Determinants of Health Assessment	Dangerous arrhythmia absent	Up to chair[D]	Advance diet - IV fluids - IV or oral medications	Temporary pacing absent	Wean parenteral vasoactive medication - Possible inhaled pulmonary vasodilators - Immunosuppressive medications - Aspirin

6	- Floor - Social Determinants of Health Assessment	Pulmonary edema absent or mild	Possible limited ambulation[D]	- Advance diet - IV or oral medications	Arterial line absent - Endomyocardial biopsy	Parenteral vasoactive medication absent - Immunosuppressive medications - Aspirin
7	- Floor - Social Determinants of Health Assessment	Peripheral edema absent or mild	Possible limited ambulation[D]	Oral hydration - Advance diet - IV or oral medications	Pacer wires absent - Endomyocardial biopsy	- Immunosuppressive medications - Aspirin
8	- Floor - Social Determinants of Health Assessment	Stable fluid balance Biopsy not indicated or stable post biopsy	Limited ambulation[D]	- Oral hydration - Oral medications - Advance diet	ECG monitoring	- Immunosuppressive medications - Aspirin
9	- Floor - Social Determinants of Health Assessment	No evidence of organ rejection	Possibly ambulatory	- Possibly begin usual diet - Oral hydration - Oral medications	Central lines absent - ECG monitoring	- Immunosuppressive medications - Aspirin
10	- Floor - Social Determinants of Health Assessment	Adequate diet tolerated	Possibly ambulatory	Usual diet - Oral hydration - Oral medications - Oral diet	ECG monitoring	- Immunosuppressive medications - Aspirin
11	- Floor - Social Determinants of Health Assessment	Oxygen saturation greater than 90% on room air Heart rate acceptable	Ambulatory[E]	- Oral hydration - Oral medications - Oral diet	Oxygen absent - ECG monitoring	- Immunosuppressive medications - Aspirin
12	- Floor - Social Determinants of Health Assessment	No evidence of postoperative or surgical site infection	Ambulatory	- Oral hydration - Oral medications - Oral diet	ECG monitoring absent	- Immunosuppressive medications - Aspirin
13	- Social Determinants of Health Assessment - Floor to discharge - Complete discharge planning	Hemodynamic stability Dangerous arrhythmia absent No evidence of postoperative or surgical site infection No evidence of organ rejection Pulmonary edema absent or mild	Ambulatory or acceptable for next level of care[E]	Oral hydration[F] Oral medications or regimen acceptable for next level of care Oral diet or acceptable for next level of care		- Immunosuppressive medications - Aspirin

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|--|--|--|--|--|--|--|
| | | <ul style="list-style-type: none"> • Immunosuppression regimen appropriate for next level of care • Cardiac function and fluid balance appropriate for next level of care • Oxygen saturation greater than 90% on room air • Pain absent or managed • Discharge plans and education understood | | | | |
|--|--|--|--|--|--|--|

(1)(4)(8)(19)(36)(37)(38)(39)

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 12 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(1)(36)(38)(39):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Severe pretransplant condition (eg, heart assist device, intra-arterial balloon, extracorporeal membrane oxygenation (ECMO))(37)
 - Expect brief to moderate stay extension.
 - Complications of procedure(10)(19)(40)
 - Hemorrhage, infection, pulmonary hypertension, nonspecific graft failure, stroke, and multiple organ failure may occur and require medical and possibly surgical management.
 - Stay extension varies.
 - Primary allograft dysfunction(41)
 - Anticipate inotropic agents, possible ventricular assist device, ECMO, or other mechanical support.
 - Retransplant may be indicated for patients who do not recover cardiac function and have low-risk factors for recurrence.
 - Expect moderate to prolonged stay extension.
 - Acute rejection
 - Anticipate endomyocardial biopsy and serum assays for donor specific and anti-HLA antibodies.
 - Patients may require cytotoxic, antibody-directed, or immunosuppressive therapy, inotropic agents, and possible ventricular assist device, ECMO, or other mechanical support.
 - Expect moderate stay extension.
 - Post-transplant development of hemodynamic instability, respiratory failure, or renal insufficiency(19)(29)(36)(42)
 - Expect brief to moderate stay extension.
 - Post-transplant arrhythmia(43)
 - Anticipate medical and device (eg, pacemaker) treatment of arrhythmia.[G]
 - Expect brief stay extension.
 - Active comorbidities
 - Patient with unstable comorbidities, such as obstructive or restrictive lung disease, pretransplant hepatic or renal insufficiency, pulmonary hypertension, diabetes mellitus, or cerebral or peripheral vascular disease, may require continued inpatient care.
 - Expect brief to moderate stay extension.
 - Transplant performed in infancy (ie, less than 1 year of age)(36)(44)
 - Expect brief to moderate stay extension.

See Common Complications and Conditions [ISC](#) for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(1)(4)(8)(19):
 - Diagnostic test scheduling and completion, including:

- Laboratory tests (eg, CBC, electrolytes, biomarkers, immunosuppressant levels)(45)
 - Imaging (eg, serial chest x-rays, myocardial perfusion imaging)(46)
 - Echocardiogram
- Treatment and procedure scheduling and completion, including:
 - IV antibiotics
 - Transfusion
 - Inotropic support
 - Right ventricular assist device support
 - Nitric oxide or inhaled prostacyclin
 - Temporary pacemaker
 - Endomyocardial biopsy
 - Thoracentesis
 - Establishment of immunosuppression protocols(47)
- Consultation, assessment, and other services scheduling and completion, including:
 - Medical and surgical specialty consultations as indicated
- Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool [SR](#)); key items include:
 - Hemodynamic stability measurements, which may include(29):
 - Blood pressure via arterial line
 - Pulmonary artery pressure
 - Central venous pressure
 - Cardiac output and cardiac index
 - Urine output
 - Need for vasoactive medications
 - Need for pulmonary vasodilator agents (eg, nitric oxide, inhaled prostacyclin)
 - Respiratory function and oxygenation
 - Need for chest tube
 - Cardiovascular status, including presence of:
 - Arrhythmias
 - Electrolyte imbalances, especially potassium and magnesium
 - Signs and symptoms of cardiac tamponade, bleeding, hypertension, or hypovolemia
 - Wound and sternum tenderness, swelling, or serous drainage from incision
 - Pain management
 - Renal function
 - Gastrointestinal status
 - Signs and symptoms of infection, including:
 - Fever greater than 100.4 degrees F (38 degrees C)
 - Hypotension
 - Elevated cardiac output and cardiac index
 - Low systemic vascular resistance
 - Elevated WBC count
 - Purulent drainage from site of infection
 - Increasing oxygen requirements
 - Mental status change

Discharge

Discharge Planning

- Discharge planning includes^[H]:
 - Assessment of needs and planning for care, including(49)(50):
 - Develop and modify treatment plan (involving multiple providers) as needed.
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment [SR](#) for further information.
 - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool [SR](#) for further information.(48)
 - Evaluate and address patient or caregiver preferences as indicated.
 - Identify skilled services needed at next level of care, with specific attention to:
 - Cardiovascular status assessment(1)
 - Medication management, adherence instruction, and side effects assessment(51)(52)(53)
 - Pain management(54)
 - Wound or dressing management
 - Early identification of anticipated discharge destination; options include(50)(55):
 - Home, considerations include:

- Home safety assessment. See Home Safety Assessment [SR](#) for further information.
- ☐ Patient safe to go home; examples include(56)(57)(58):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
- Access to follow-up care
- Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#) for further information.
- Caregiver need, ability, and availability
- Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool [SR](#) for further information.
- Transitions of care plan complete, including(50)(55)(59):
 - Patient and caregiver education complete. See Heart Transplant, Pediatric: Patient Education for Clinicians [SR](#) for further information.
 - See Teach Back Tool [SR](#) for further information.
 - ☐ Medication reconciliation completion includes(53)(60):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(61)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See Medication Reconciliation Tool [SR](#) for further information.
 - Plan communicated to patient, caregiver, and all members of care team, including(62):
 - Inpatient care and service providers
 - Primary care provider
 - All post-discharge care and service providers
 - Appointments planned or scheduled, which may include(63):
 - Primary care provider
 - Cardiologist
 - Cardiothoracic surgeon
 - Rehabilitation therapy services(1)
 - Specialists for management of comorbidities as needed(63)
 - Transplant center or team
 - Other
 - Outpatient testing and procedure plans made, which may include:
 - Laboratory testing
 - Radiology
 - Other
 - Referrals made for assistance or support, which may include:
 - Community services(64)
 - Educational program (eg, chronic condition management, self-management)(65)
 - Financial, for follow-up care, medication, and transportation
 - Self-help or support groups
 - Tobacco use treatment
 - Other
 - Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
 - Blood pressure cuff
 - Child safety seat(66)

- Nutritional supplements
- Weight scale
- Wound care equipment and supplies(67)
- Other

Discharge Destination

- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See Home Care Indications for Admission Section [HC](#) in Transplant, Pediatric guideline in Home Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 13 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, and 3 book sections.

Heart-lung transplant may be performed in patients with irreversible pulmonary hypertension, certain congenital heart diseases, Eisenmenger syndrome not amenable to surgical correction or medical therapy, or severe lung disease in conjunction with irreversible myocardial dysfunction or irreparable congenital defects.(6)(7) **(EG 2)** Heart-liver transplant may be performed for patients with advanced heart failure and concomitant cirrhosis, Fontan circulation with severe Fontan-associated liver disease, selected patients with cardiac and hepatic amyloidosis, or other associated liver disease that may affect the cardiac allograft.(8)(9) **(EG 2)** Heart-kidney transplant may be performed for patients with advanced heart failure and concomitant renal failure.(9) **(EG 2)**

Alternatives

Patients with advanced heart failure can show improvement via maximizing medical therapy and, when applicable, using cardiac resynchronization therapy, implantable cardioverter-defibrillators, revascularization, ventricular resection, or valve repair or replacement.(1)(2)(13) **(EG 2)**

Length of Stay

Analysis of a database including 1145 pediatric patients undergoing heart transplant for myocarditis or cardiomyopathy found that 25% were discharged within 12 days.(37) **(EG 2)** Analysis of national hospital discharge data for a pediatric population shows 24% of hospitalized patients undergoing the principal procedure of heart transplant discharged in 12 days or less.(38) **(EG 3)**

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Footnotes

- [A] Peak metabolic oxygen consumption should be measured on patients receiving maximally tolerated guideline directed medical therapy (GDMT).(14) [A in Context Link 1]
- [B] A Fontan circulation is created surgically, usually as palliation for single ventricular congenital heart disease.(18) Failed Fontan circulation may be an indication for combined heart-liver transplant.(18) [B in Context Link 1]
- [C] See Clinical Indications for Procedure in this guideline. [C in Context Link 1]
- [D] For very young patients (eg, infants), activity goals should be modified based on age and development. [D in Context Link 1, 2, 3, 4, 5]
- [E] Patient is ambulatory or near baseline activity for age and development. [E in Context Link 1, 2]
- [F] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [F in Context Link 1]
- [G] Permanent pacemaker following heart transplant is indicated for persistent bradycardia that does not improve spontaneously and is either symptomatic or limits rehabilitation, or for atrioventricular block due to graft vasculopathy.(43) [G in Context Link 1]
- [H] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(48) [H in Context Link 1]

Definitions

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

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Dangerous arrhythmia

- Dangerous arrhythmia, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Acute myocarditis
 - Myocardial ischemia
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Heart rhythms of concern due to **1 or more** of the following:
 - Hypotension
 - Respiratory distress

- Association with other significant symptoms (eg, syncope)(9)(10)(11)

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Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older[A]
 - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

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Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

No contraindications to heart transplant present

- No contraindications to heart transplant present,[A] as indicated by **ALL** of the following(1)(3)(4)(5)(6)(7)(8)(9)(10)(11):
 - No severe systemic illness limiting life expectancy to less than 2 years (eg, malignancy, AIDS, autoimmune disease, Duchenne muscular dystrophy)(11)(12)(13)
 - No congenital anomaly that would preclude successful cardiac transplant (eg, severe pulmonary artery hypoplasia, severe pulmonary vein stenosis or aplasia)(14)
 - No irreversible severe hepatic dysfunction if considered for only heart transplant (eg, INR greater than 1.5 off warfarin)
 - No severe renal insufficiency if considered for only heart transplant (eg, GFR less than 30 mL/min/1.73m² (0.5 mL/sec/1.73m²))(10)(11)
 - No severe obstructive pulmonary disease (eg, FEV₁ less than 1 L/minute)[B]
 - No severe fixed (ie, pharmacologically irreversible) pulmonary hypertension (may include pulmonary veno-occlusive disease)[B](13)
 - No psychiatric illness or severe cognitive disability (eg, dementia) that may interfere with postoperative medical adherence
 - No severe neurologic disorder inhibiting postoperative recovery (eg, severe amyloid peripheral neuropathy, severe dysautonomia)(11)
 - No active substance abuse[C][D](2)(15)
 - No active infection (except device-related infection in ventricular-assist device recipients)
 - No clinically severe symptomatic cerebrovascular disease
 - No active peptic ulcer disease
 - No other potentially reversible contraindications (eg, poorly controlled severe hypertension)

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Footnotes

- A. Specialty guidelines and narrative reviews suggest that right heart catheterization be performed every 3 to 6 months to reassess pertinent findings.(1)(2)
- B. This contraindication may not be applicable to patients undergoing a combined heart-lung transplant.
- C. Recipient eligibility regarding substance abuse may vary between transplant programs and may be based on health insurer requirements as well as state legislation.(15)
- D. A structured drug or alcohol rehabilitative program may be appropriate in patients with a recent history of alcohol or drug abuse.(2)

Orthostatic hypotension

- Orthostatic hypotension,^{[A][B]} as indicated by **1 or more** of the following(1)(2)(3):
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

1. Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing Group. Evaluation and treatment of orthostatic hypotension. *Journal of the American Society of Hypertension* 2013 Jul-Aug;7(4):317-324. DOI: 10.1016/j.jash.2013.04.006.
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3. Fang JC, O'Gara PT. History and physical examination: an evidence-based approach. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. *Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine*. 12th ed. Elsevier; 2022:123-140.

Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
 - Patient with **1 or more** of the following:
 - Dyspnea (difficulty breathing)
 - Tachypnea
 - Abnormal breathing pattern (eg, chest retractions)
 - Other evidence of difficulty breathing
 - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
 - Hypoxemia
 - Altered mental status
 - Other evidence of respiratory compromise (eg, respiratory acidosis)

References

1. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
2. Naureckas ET, Solway J. Disturbances of respiratory function. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2133-2139.

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following(5):
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents
 - Smoke detectors not working or unavailable
 - Food insecurity, as indicated by **1 or more** of the following(6):
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
 - Insufficient transportation, as indicated by **1 or more** of the following(7):
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
 - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
 - Insufficient utilities, as indicated by **1 or more** of the following(8):
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
 - Personal safety risk, as indicated by **2 or more** of the following(6):
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
 - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
 - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
 - Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
 - Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

References

1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
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3. Daniel-Robinson L, Moore JE. Innovation and Opportunities to Address Social Determinants of Health in Medicaid Managed Care. [Internet] Institute for Medicaid Innovation. 2019 Jan Accessed at: <https://medicaidinnovation.org/emerging-topics/>. [accessed 2023 Oct 20]
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Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following(1)(2):
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. *Principles and Practice of Hospital Medicine*. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. *The Harriet Lane Handbook: A Manual for Pediatric House Officers*. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,^{[A][B]} as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Greater than 18 breaths per minute in adult or child 13 years of age or older^[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age^[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age^[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age^[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age^[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age^[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age^[A]

References

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3. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. *The Harriet Lane Handbook: A Manual for Pediatric House Officers*. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.⁽¹⁾ This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.⁽¹⁾

Codes

ICD-10 Diagnosis: C38.0, D15.1, I09.81, I42.0, Q23.4, Q24.8, T86.22, T86.290, T86.32

ICD-10 Procedure: 02YA0Z0, 02YA0Z1, 02YA0Z2

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